



Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Lab 1 Solution

Full Name: LAIBAH BAKHTAWAR

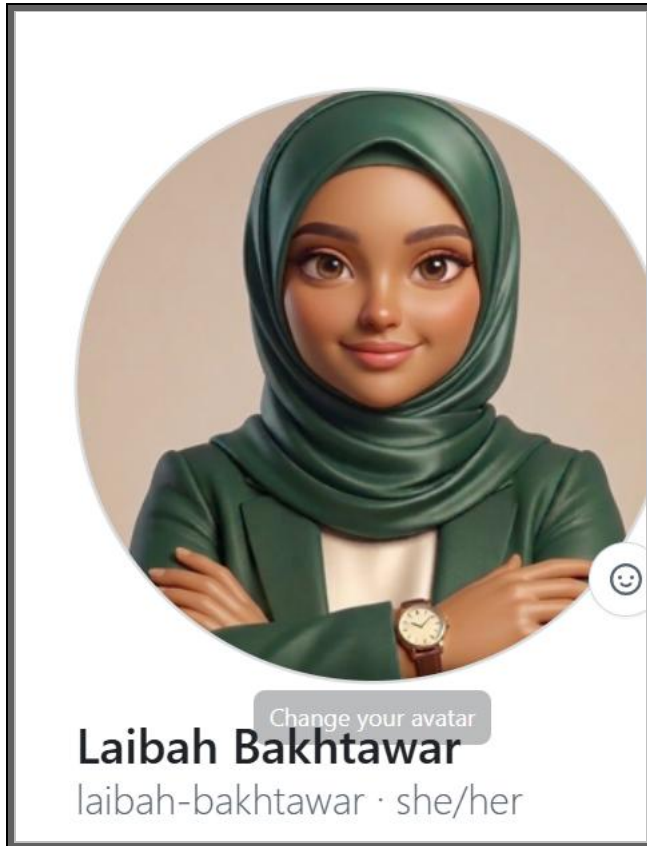
Registration Number: FA24B1-SE-031

Course Code: BSE-665

Section: A

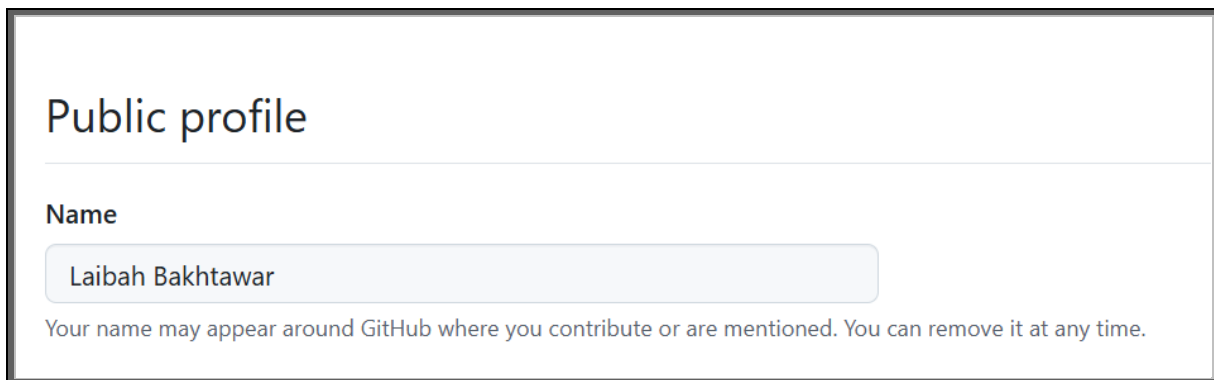
GitHub Username: laibah-bakhtawar

Task 1: Create and Configure Your GitHub Account



File: *github_profile.png*

Caption: GitHub profile page clearly showing the profile name and GitHub username.



File: *github_actual_name.png*

Caption: GitHub Settings > Public profile page showing the complete actual name entered in the Name field.

Task 2: Register in the Student Marks Portal

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

For example: 2024-BSE-01

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

Select course

Section

Select section

Submit enrollment

Sign out

File: *portal_login_page.png*

Caption: Student Marks Portal login page before signing in.



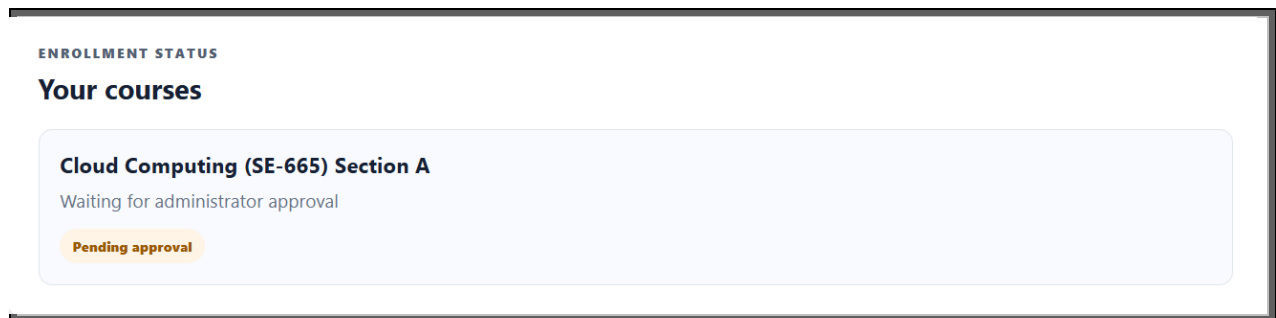
Laibah Bakhtawar

2024-BSE-31

+ Enroll in another course

File: *portal_github_profile.png*

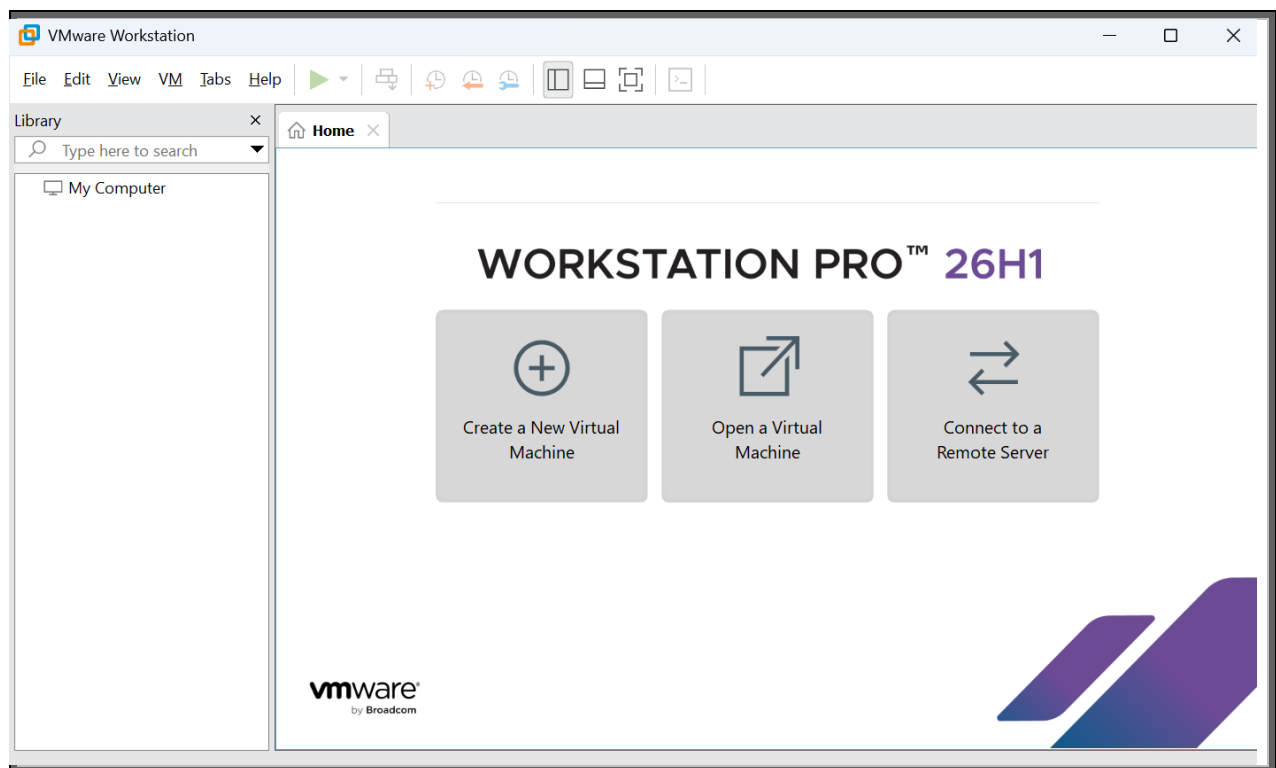
Caption: GitHub profile information displayed inside the Student Marks Portal after signing in with GitHub.



File: *portal_enrollment_submitted.png*

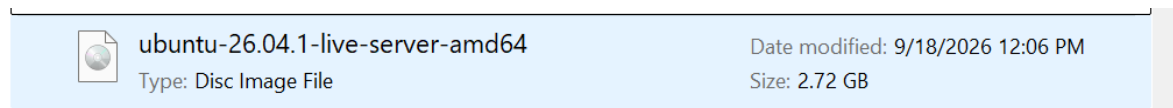
Caption: Submitted enrollment showing registration number, course code, and section as pending/approved.

Task 3: Download VMware Workstation Pro and Ubuntu Server



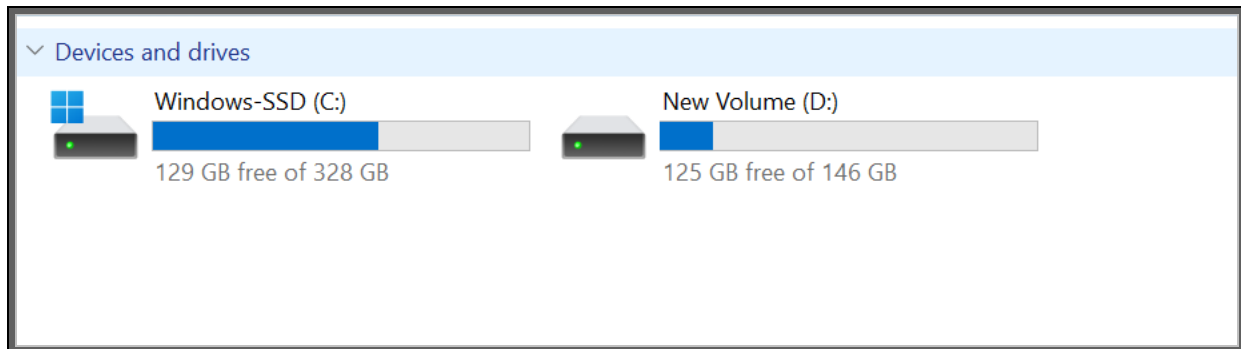
File: *vmware_workstation.png*

Caption: VMware Workstation Pro running successfully on the local machine.



File: *ubuntu_server_iso.png*

Caption: Downloaded Ubuntu Server LTS ISO file showing its filename.



File: *available_storage.png*

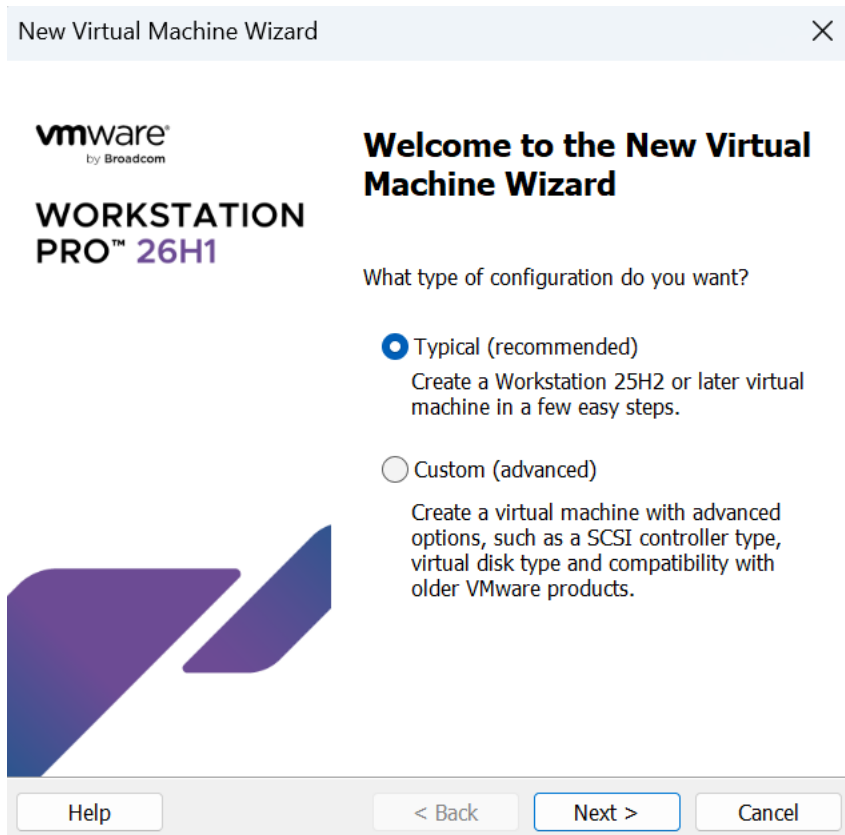
Caption: Available disk storage showing at least 20 GB free for the virtual machine.

Task 4: Create the Ubuntu Server Virtual Machine



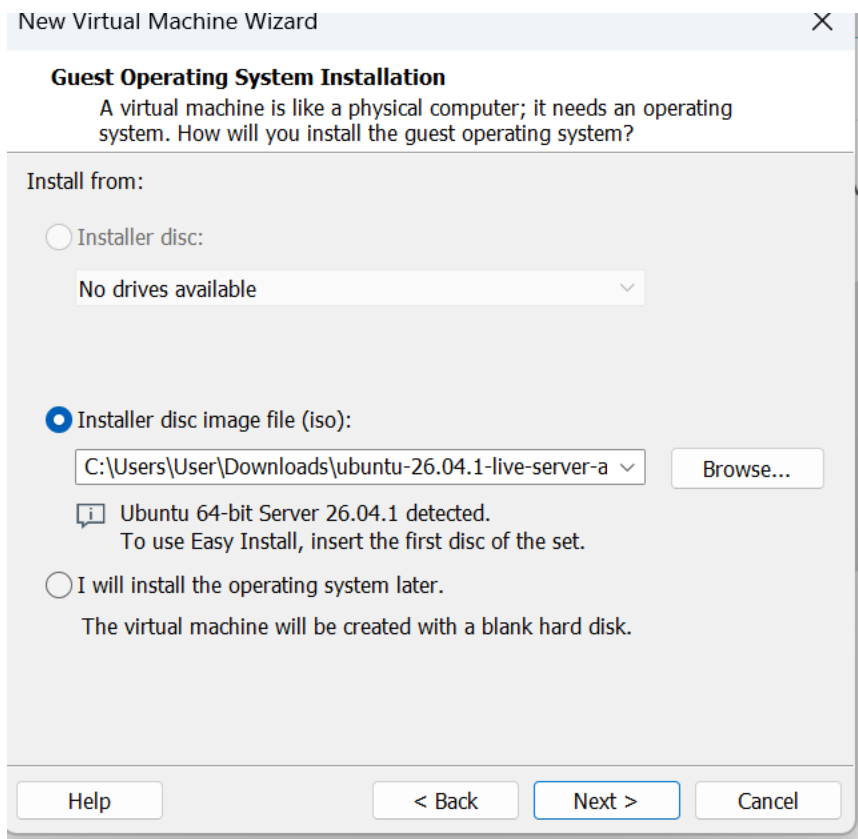
File: *vm_creation_wizard.png*

Caption: New Virtual Machine Wizard opened in VMware Workstation Pro.



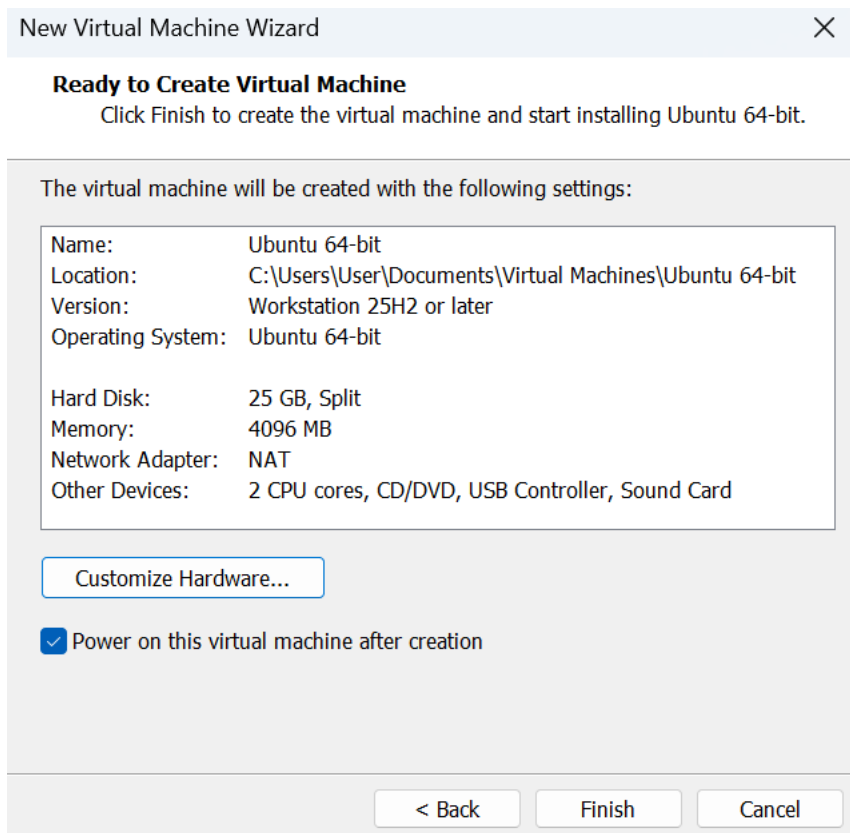
File: *vm_typical_configuration.png*

Caption: Typical (recommended) configuration option selected in the wizard.



File: *vm_ubuntu_iso_selected.png*

Caption: Installer disc image file (ISO) option with the downloaded Ubuntu Server ISO path selected.



File: *vm_configuration_summary.png*

Caption: Final virtual machine configuration summary with at least 20 GB of allocated disk space.

```

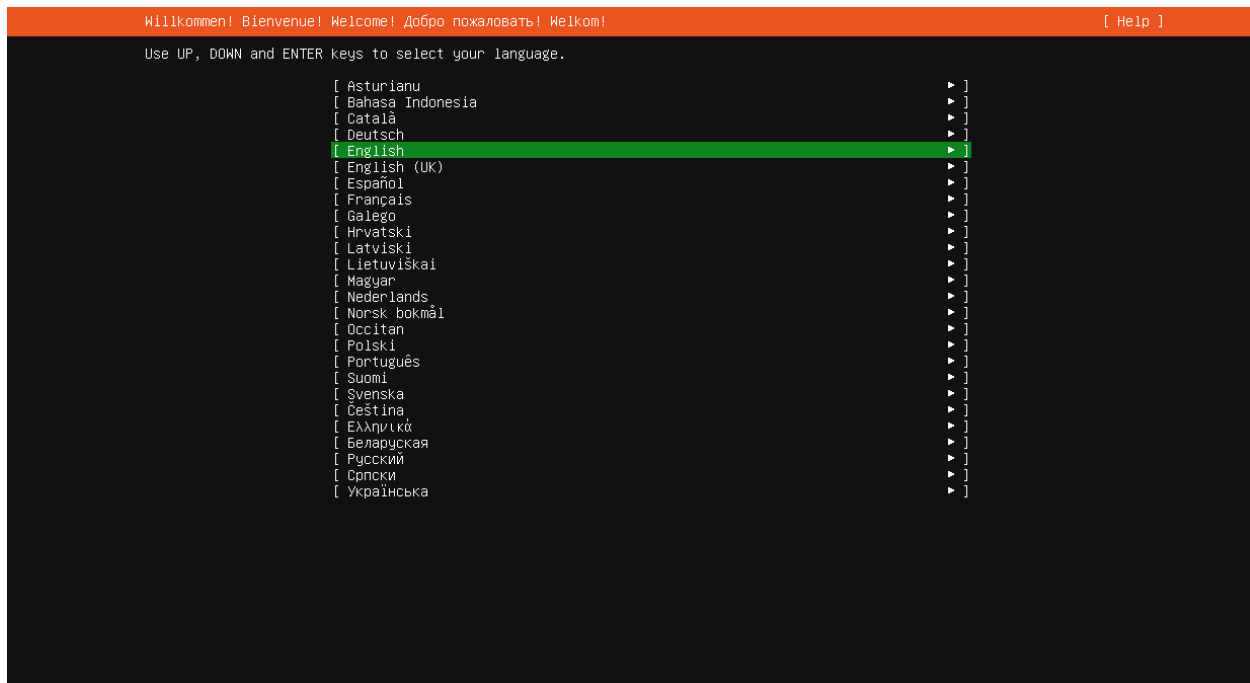
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished finalrd.service - Create runtime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Plymouth To Write Out Runtime Data.
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Finished netplan-config.service - Netplan Backend Configuration.
Mounting proc-sys-fs-binfmt_misc.mount - Executable File Formats File System...
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - Executable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog-update.service - Rebuild Journal Catalog...
[ OK ] Started vgauth.service - Authenticate virtual machines hosted on VMware.
[ OK ] Started open-vm-tools.service - Set up virtual machines hosted on VMware.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
[ OK ] Finished systemd-udev-trigger.service - Coldplug All udev Devices.
[ OK ] Started systemd-ask-password-console-askq.service - Console Directory Watch.
[ OK ] Reached target cryptsetup.target - Local Encrypted Volumes.
[ OK ] Finished systemd-journal-catalog-update.service - Rebuild Journal Catalog.
[ OK ] Finished snapd.apparmor.service - Profiles managed internally by snapd.
[ OK ] Finished console-setup.service - Set console font and keymap.
[*] (1 of 2) Job cloud-init-main.service/start running (7s / no limit)
[ OK ] Listening on systemd-rfkill.socket - Load/Save RF Kill Switch Status /dev/rfkill Watch.

```

File: *ubuntu_installer_boot.png*

Caption: Ubuntu Server boot screen after powering on the virtual machine.

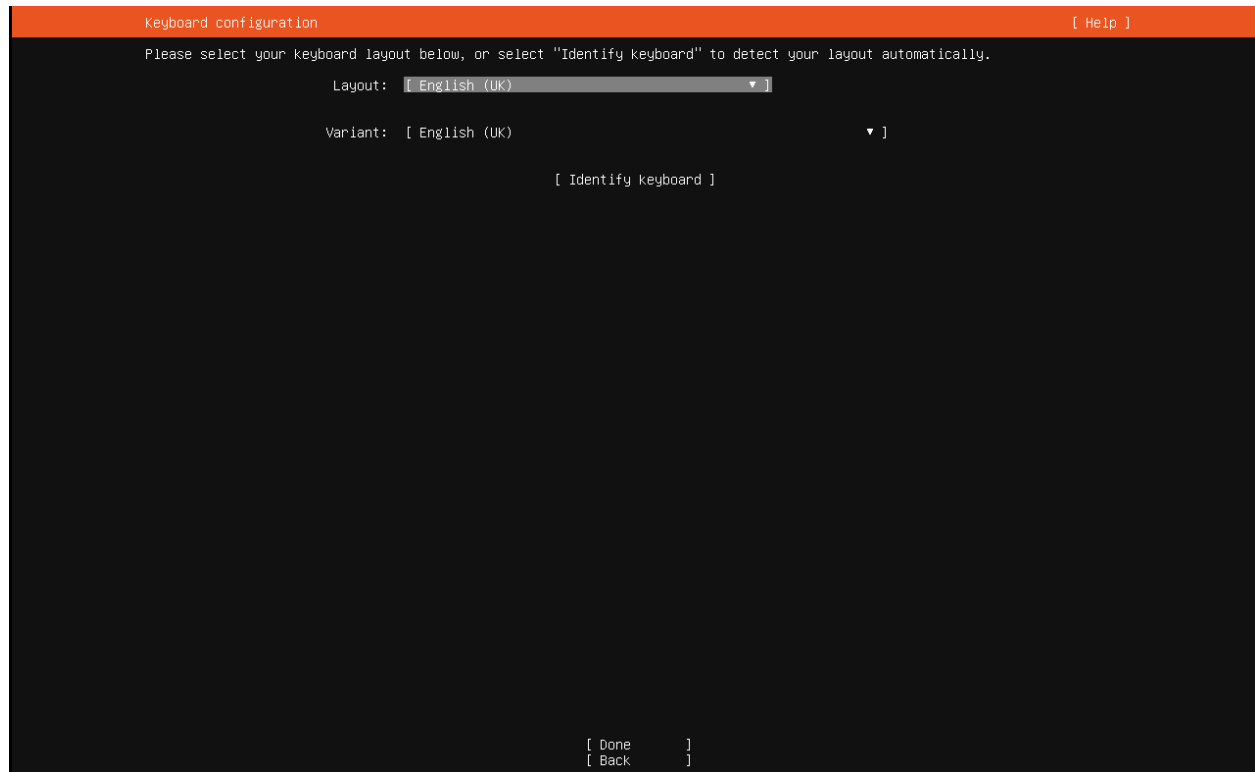
Task 5: Choose the Installation Language



File: *ubuntu_language.png*

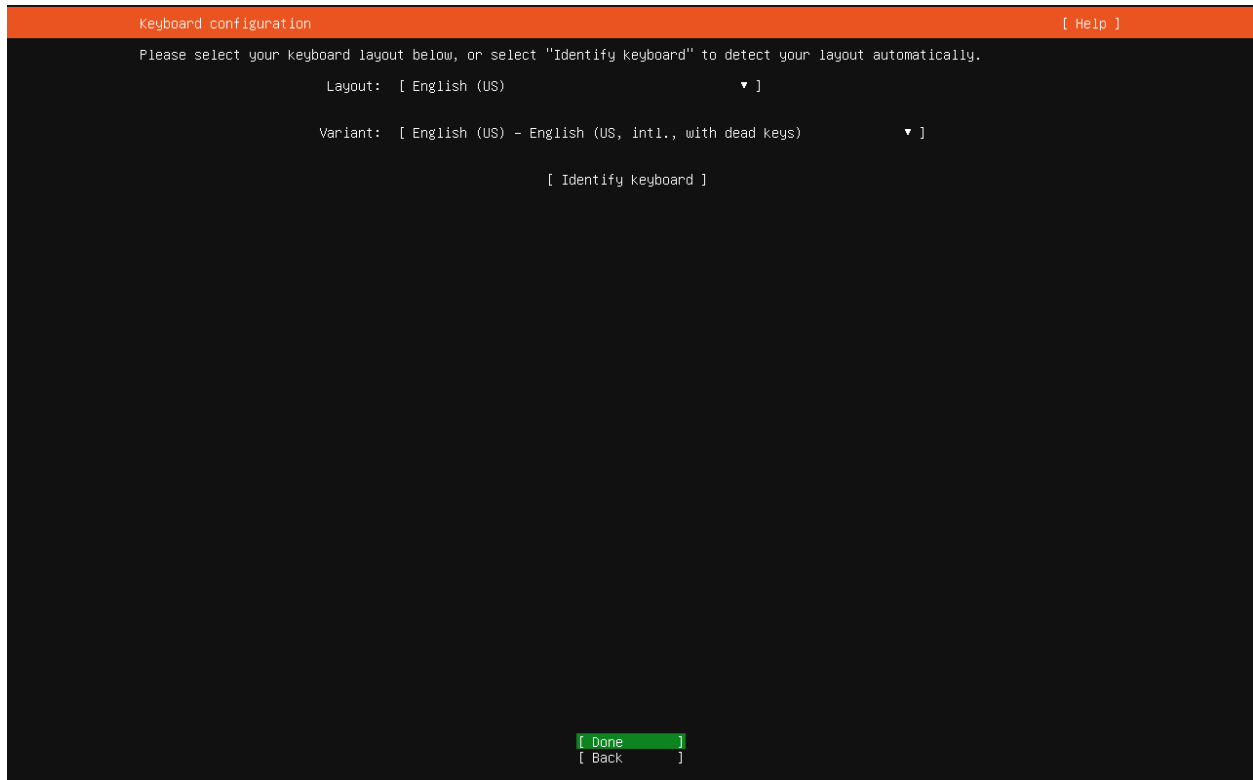
Caption: Installer screen showing the selected installation language.

Task 6: Configure the Keyboard Layout



File: *ubuntu_keyboard_layout.png*

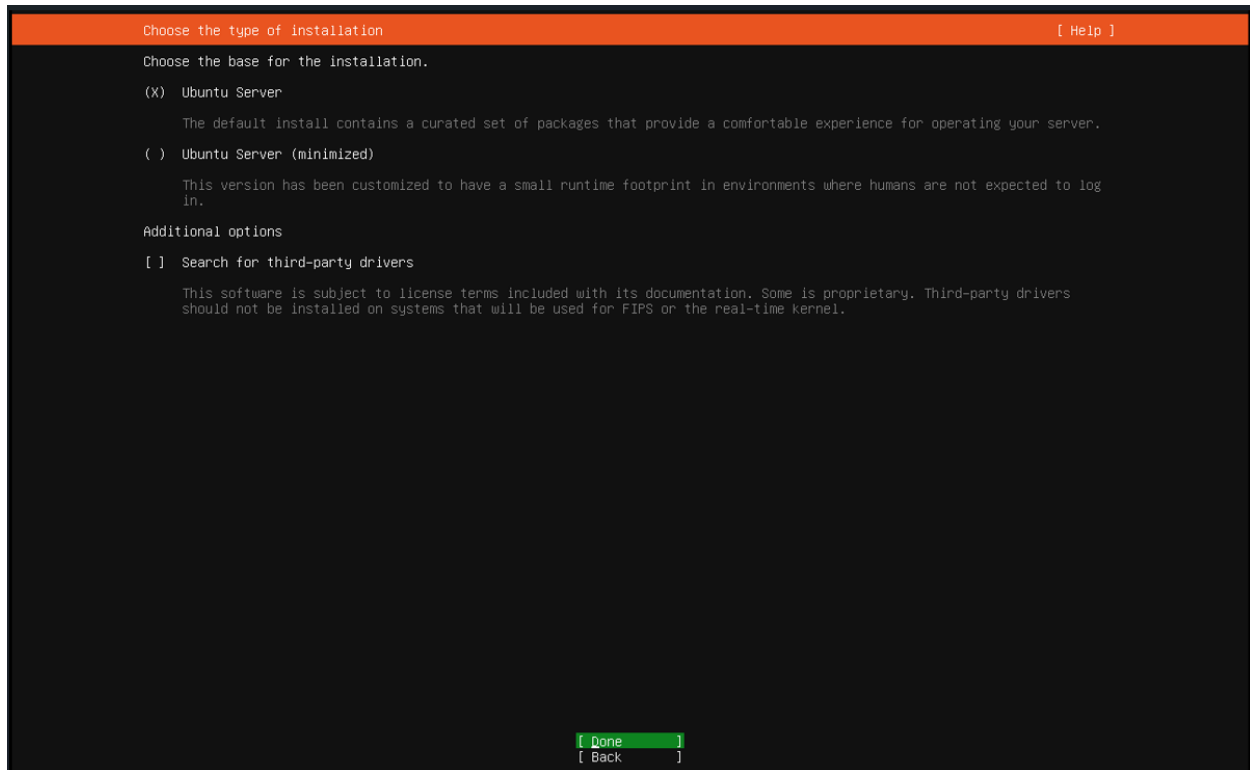
Caption: Installer screen showing the selected keyboard layout.



File: *ubuntu_keyboard_variant.png*

Caption: Installer screen showing the selected keyboard variant.

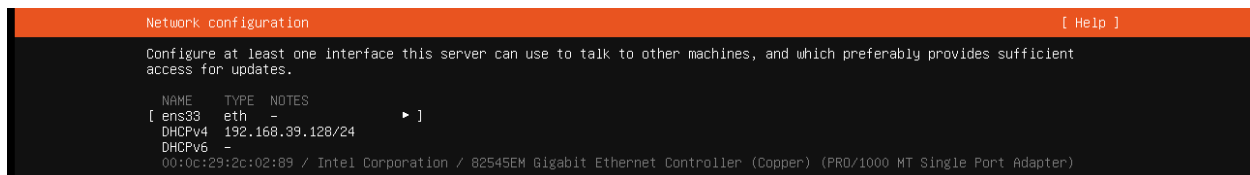
Task 7: Choose the Ubuntu Server Installation Type



File: *ubuntu_installation_type.png*

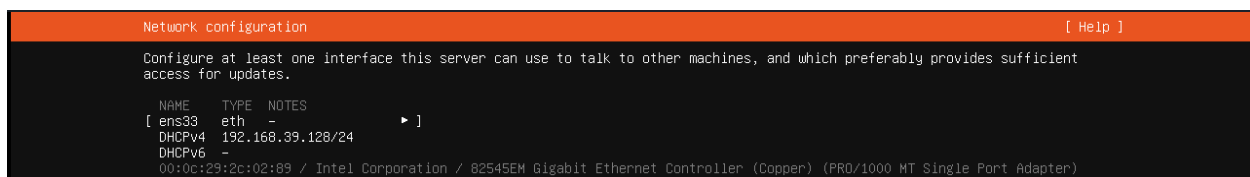
Caption: Normal Ubuntu Server installation option selected (not MAAS).

Task 8: Configure the Network



File: *ubuntu_network_interface.png*

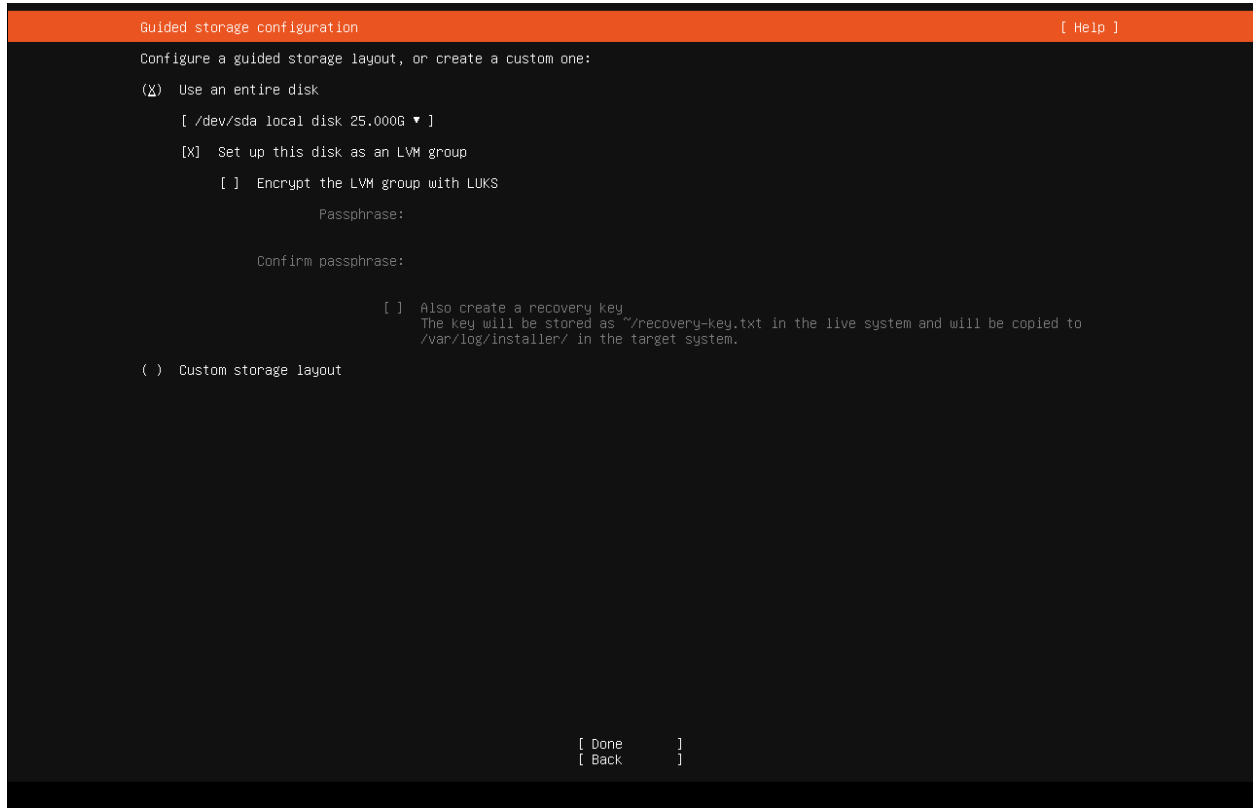
Caption: Network interface detected by the Ubuntu Server installer.



File: *ubuntu_installer_network.png*

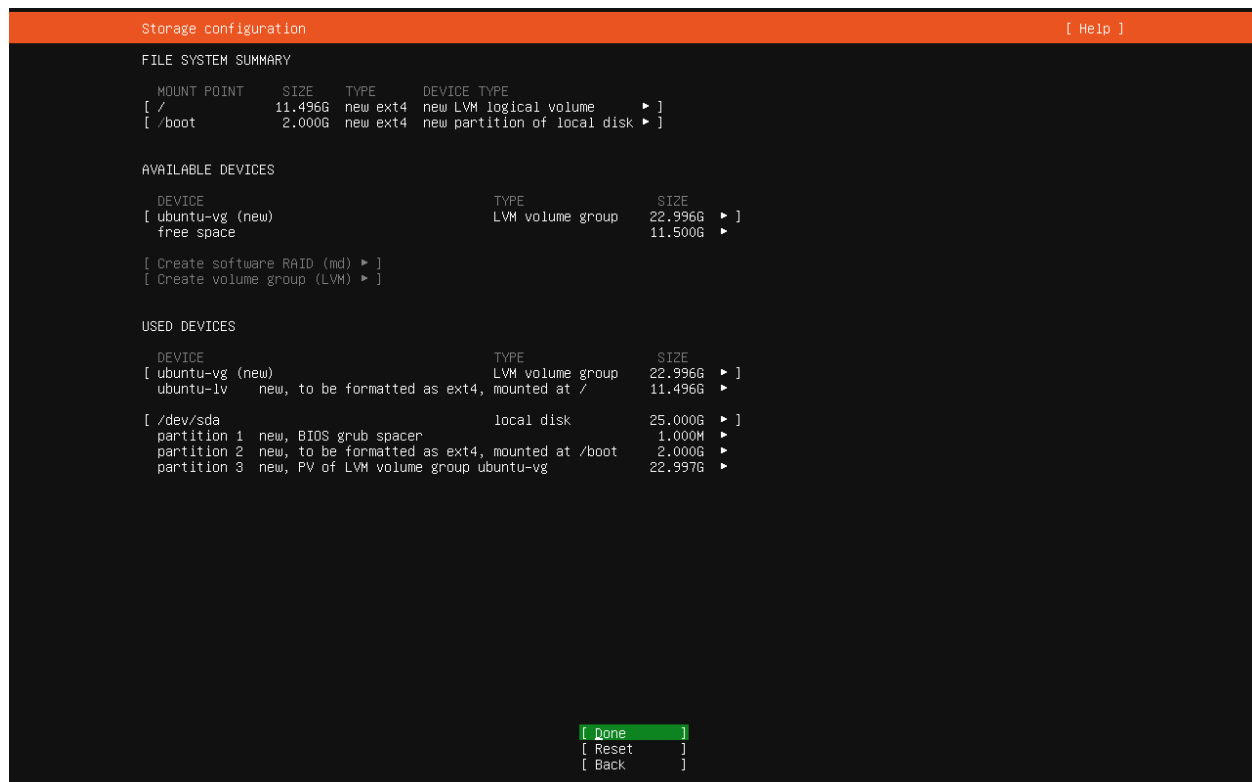
Caption: Network interface showing an IP address automatically assigned via DHCP.

Task 9: Configure Storage



File: *ubuntu_guided_storage.png*

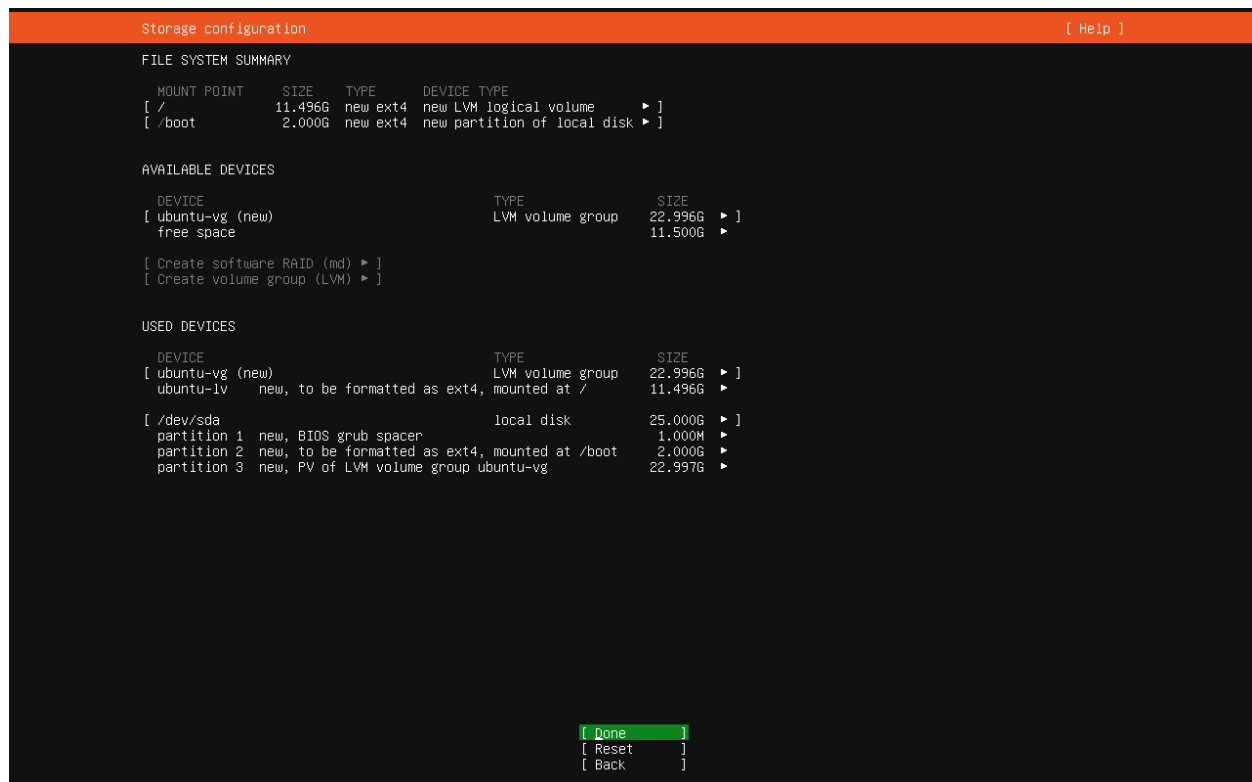
Caption: Guided storage configuration option selected to use the VMware virtual disk.



File: *ubuntu_storage_configuration.png*

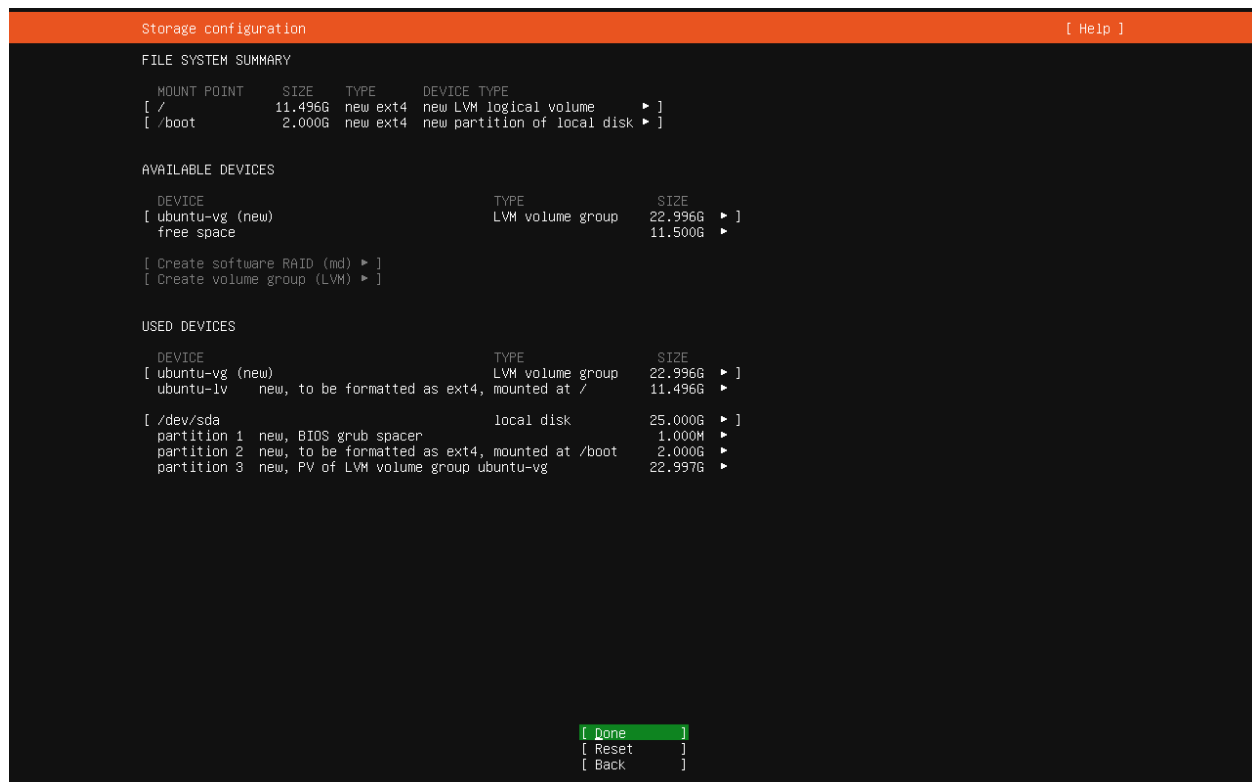
Caption: Storage configuration screen reviewed before continuing.

Task 10: Select the Installation Disk



File: *ubuntu_selected_disk.png*

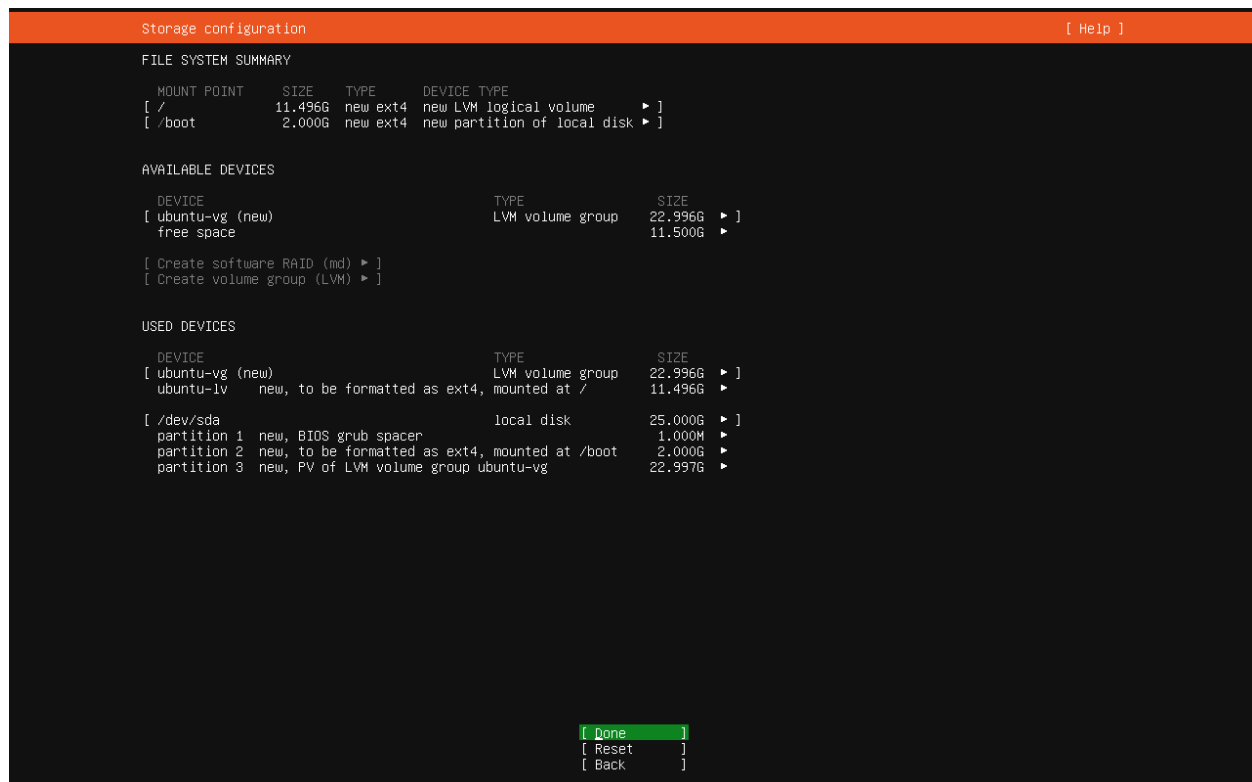
Caption: Virtual disk created for the Ubuntu Server VM selected as the installation target.



File: *ubuntu_disk_details.png*

Caption: Disk name and size confirming the correct VMware virtual disk.

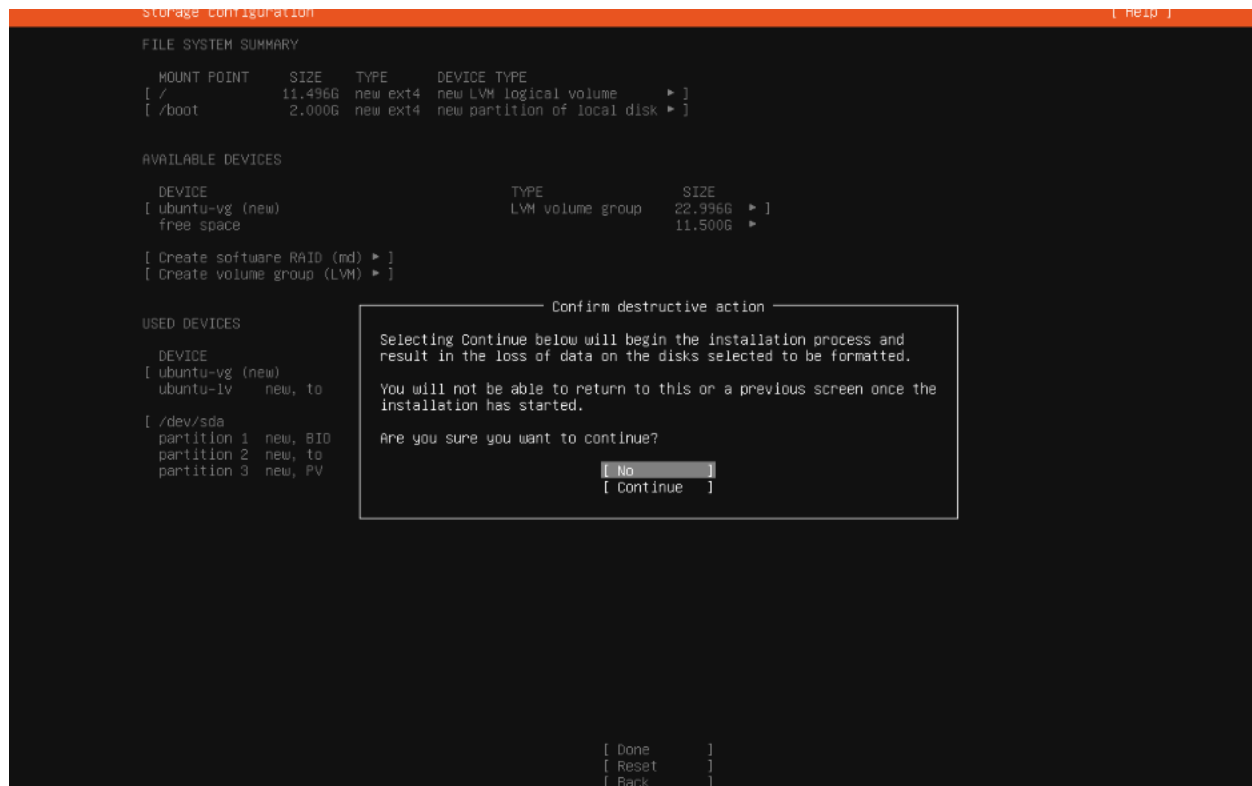
Task 11: Review the Partitions



File: *ubuntu_partition_layout.png*

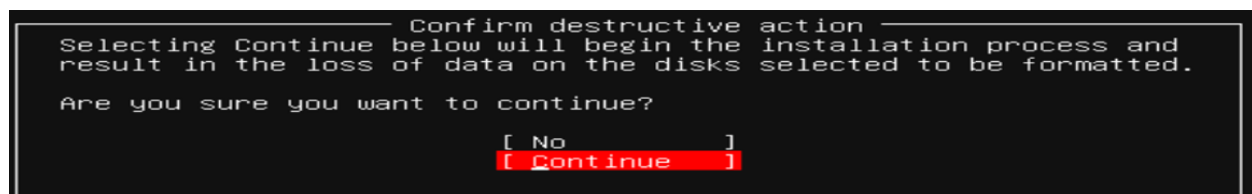
Caption: Proposed partition layout on the VMware virtual disk.

Task 12: Confirm the Storage Changes



File: *ubuntu_storage_warning.png*

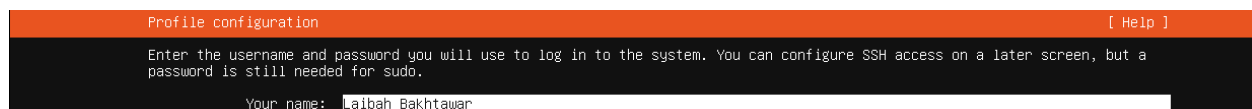
Caption: Destructive-action warning screen shown before confirming storage changes.



File: *ubuntu_storage_confirmed.png*

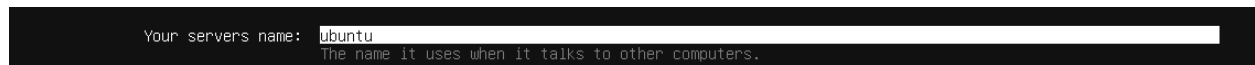
Caption: Confirmation of storage changes on the correct VMware virtual disk.

Task 13: Configure the Ubuntu Profile



File: *ubuntu_actual_name.png*

Caption: Ubuntu profile setup showing the complete actual name entered in the 'Your name' field.



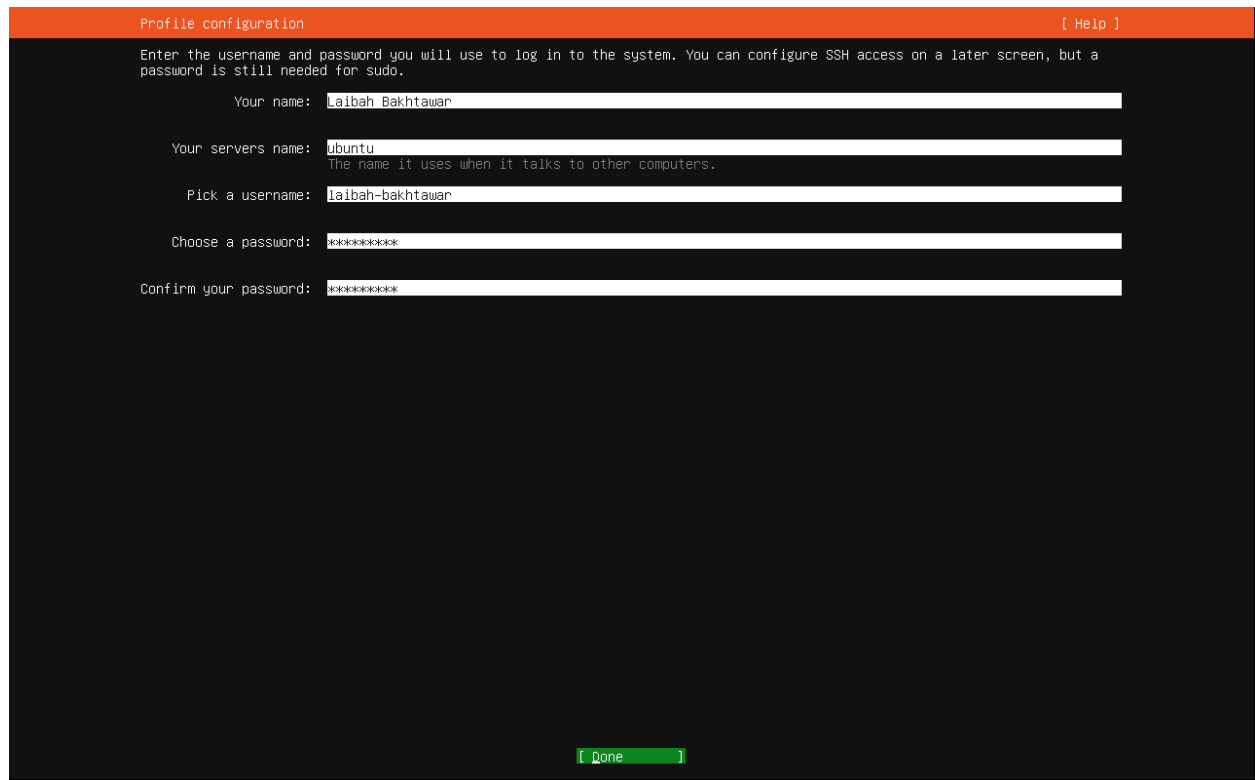
File: *ubuntu_server_name.png*

Caption: Ubuntu profile setup showing 'ubuntu' entered in the 'Your server's name' field.



File: *ubuntu_username.png*

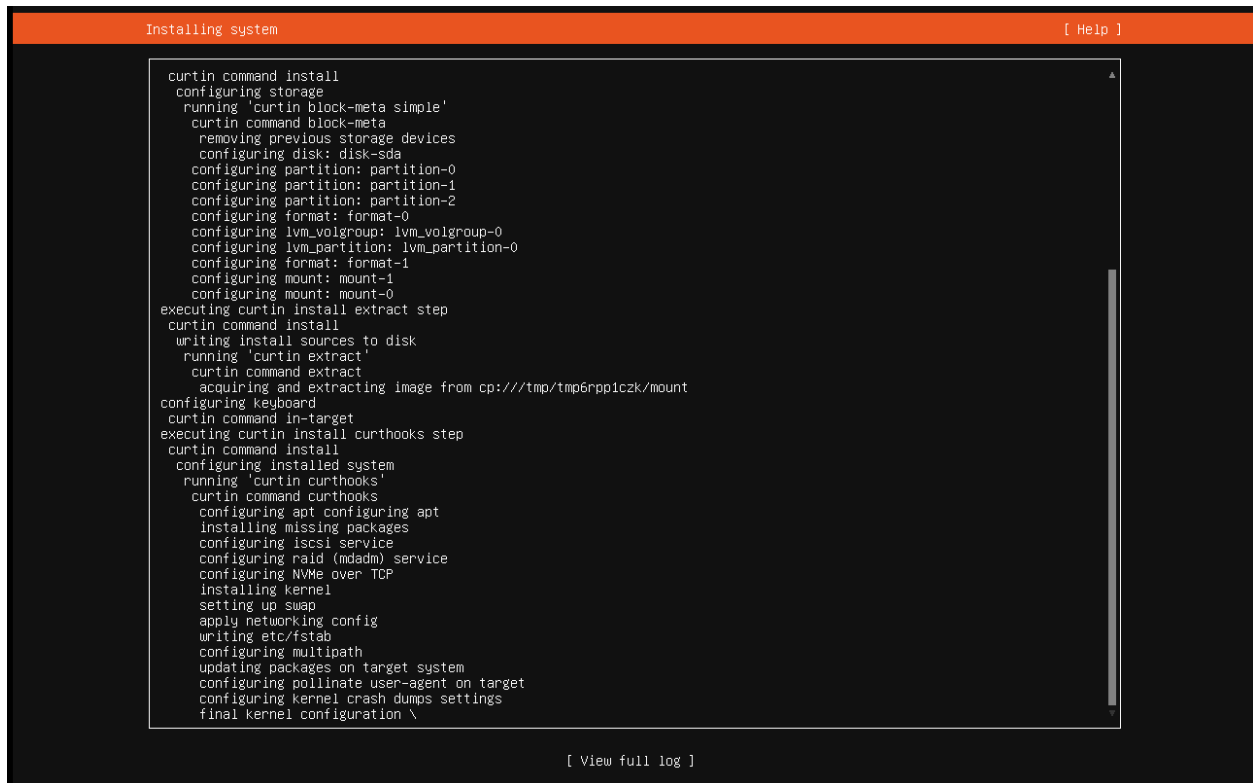
Caption: Ubuntu profile setup showing the exact GitHub username entered in the 'Pick a username' field.



File: *ubuntu_profile_setup.png*

Caption: Completed Ubuntu profile page with all required fields filled in.

Task 14: Complete the Software Installation

A screenshot of the 'Installing system' window in the Ubuntu Server installer. The window has an orange header bar with the text 'Installing system' on the left and '[Help]' on the right. The main area is dark gray and contains a white text log of the installation process. The log shows various steps such as configuring storage, partitioning disks, formatting partitions, mounting, and installing the system. At the bottom of the log area, there is a link '[View full log]'.

```
curtin command install
configuring storage
  running 'curtin block-meta simple'
  curtin command block-meta
    removing previous storage devices
    configuring disk: disk-sda
    configuring partition: partition-0
    configuring partition: partition-1
    configuring partition: partition-2
    configuring format: format-0
    configuring lvm_volgroup: lvm_volgroup-0
    configuring lvm_partition: lvm_partition-0
    configuring format: format-1
    configuring mount: mount-1
    configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmp6rppic2k/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration \
```

[View full log]

File: *ubuntu_installation_progress.png*

Caption: Ubuntu Server software installation in progress.

Task 15: Reboot and Verify the Ubuntu Identity

```
Installation complete! [ Help ]

Full installer output
Running command ['mount', '--bind', '/sys', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '--bind', '/target/usr/bin/true', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['unshare', '--help'] with allowed return codes [0] (capture=True)
Checking if target_proc (/target/proc) is a mount
It is, so unshare will use --mount-proc=/target/proc
Running command ['unshare', '--fork', '--pid', '--mount-proc=/target/proc', '--', 'chroot', '/target', 'apt-get', 'update'] with allowed return codes [0] (capture=False)
Ign:1 file:/cdrom/resolute InRelease
Get:2 file:/cdrom/resolute Release [664 B]
Get:2 file:/cdrom/resolute Release [664 B]
Hit:4 http://pk.archive.ubuntu.com/ubuntu/resolute InRelease
Get:5 http://pk.archive.ubuntu.com/ubuntu/resolute-updates InRelease [137 kB]
Hit:6 http://security.ubuntu.com/ubuntu/resolute-security InRelease
Get:7 http://security.ubuntu.com/ubuntu/resolute-security/main Translation-en [113 kB]
Hit:8 http://pk.archive.ubuntu.com/ubuntu/resolute-backports InRelease
Get:9 http://pk.archive.ubuntu.com/ubuntu/resolute/main Translation-en [524 kB]
Get:10 http://security.ubuntu.com/ubuntu/resolute-security/restricted Translation-en [77.7 kB]
Get:11 http://pk.archive.ubuntu.com/ubuntu/resolute/restricted Translation-en [25.8 kB]
Get:12 http://pk.archive.ubuntu.com/ubuntu/resolute/universe Translation-en [6329 kB]
Get:13 http://security.ubuntu.com/ubuntu/resolute-security/universe Translation-en [54.5 kB]
Get:14 http://security.ubuntu.com/ubuntu/resolute-security/multiverse Translation-en [2844 B]
Get:15 http://pk.archive.ubuntu.com/ubuntu/resolute/multiverse Translation-en [127 kB]
Get:16 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 Packages [616 kB]
Get:17 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/main Translation-en [148 kB]
Get:18 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/restricted Translation-en [80.0 kB]
Get:19 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/universe amd64 Packages [294 kB]
Get:20 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/universe Translation-en [93.7 kB]
Get:21 http://pk.archive.ubuntu.com/ubuntu/resolute-updates/multiverse Translation-en [3172 B]
Get:22 http://pk.archive.ubuntu.com/ubuntu/resolute-backports/universe Translation-en [7988 B]
Fetched 8634 kB in 13s (675 kB/s)
Reading package lists...
Running command ['udevadm', 'settle'] with allowed return codes [0] (capture=False)
TIMED subp(['udevadm', 'settle']): 0.009
Running command ['mount', '--make-private', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/usr/bin/ischroot'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/sys'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/run'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/proc'] with allowed return codes [0] (capture=False)
Running command ['mount', '--make-private', '/target/dev'] with allowed return codes [0] (capture=False)
Running command ['mount', '/target/dev'] with allowed return codes [0] (capture=False)
finish: cmd-in-target: SUCCESS: curtin command in-target

[ Close ]
```

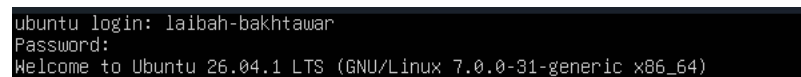
File: *ubuntu_installation_complete.png*

Caption: Installation-complete screen with the Reboot Now option.



File: *ubuntu_login_screen.png*

Caption: Ubuntu Server login screen after reboot.



File: *ubuntu_terminal_login.png*

Caption: Successful terminal login using the configured Ubuntu username and password.

```
ubuntu login: laibah-bakhtawar
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 08:47:53 AM UTC 2026

System load:  0.12           Processes:            251
Usage of /:   47.2% of 11.21GB Users logged in:             0
Memory usage: 11%           IPv4 address for ens33: 192.168.39.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibah-bakhtawar@ubuntu:~$
```

File: *ubuntu_identity_verified.png*

Caption: Terminal prompt clearly showing the required <github-username>@ubuntu:~\$ format.

Task 16: Find the Ubuntu Server IP Address

```
laibah-bakhtawar@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:2c:02:89 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c292c0289
    inet 192.168.39.129/24 metric 100 brd 192.168.39.255 scope global dynamic ens33
        valid_lft 1422sec preferred_lft 1422sec
    inet6 fe80::20c:29ff:fe2c:289/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

File: *ubuntu_ip_addr_command.png*

Caption: Output of the 'ip addr' command run inside Ubuntu Server.

```
laibah-bakhtawar@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:2c:02:89 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c292c0289
    inet 192.168.39.129/24 metric 100 brd 192.168.39.255 scope global dynamic ens33
        valid_lft 1422sec preferred_lft 1422sec
    inet6 fe80::20c:29ff:fe2c:289/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
```

File: *ubuntu_ip_address.png*

Caption: Correct IPv4 address (not 127.0.0.1) shown with the <github-username>@ubuntu prompt visible.

Task 17: Connect to Ubuntu Server from Windows

```
bash: syntax error near unexpected token `newline'
laibah-bakhtawar@ubuntu:~$ ssh laibah-bakhtawar@192.168.39.129
```

File: *windows_ssh_command.png*

Caption: SSH command run from Windows Command Prompt/PowerShell using the Ubuntu username and server IP.

```
laibah-bakhtawar@ubuntu:~$ ssh laibah-bakhtawar@192.168.39.129
The authenticity of host '192.168.39.129 (192.168.39.129)' can't be established.
ED25519 key fingerprint is: SHA256:zZCHPnN36I+Q+mx40ognvj1TY08wQWtTOghbDoVF3c
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.39.129' (ED25519) to the list of known hosts.
```

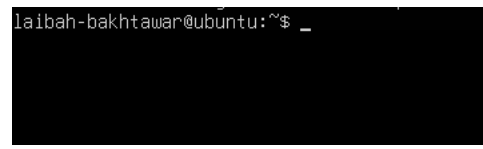
File: *windows_ssh_fingerprint.png*

Caption: Server fingerprint prompt accepted by typing 'yes'.

```
laibah-bakhtawar@ubuntu:~$ ssh laibah-bakhtawar@192.168.39.129
The authenticity of host '192.168.39.129 (192.168.39.129)' can't be established.
ED25519 key fingerprint is: SHA256:zZCHPnN36I+Q+mx40ognvj1TY08wQWtTOghbDoVF3c
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.39.129' (ED25519) to the list of known hosts.
laibah-bakhtawar@192.168.39.129's password:
Connection closed by 192.168.39.129 port 22
laibah-bakhtawar@ubuntu:~$
```

File: *windows_ssh_login.png*

Caption: Successful SSH login session into the Ubuntu Server.



```
laibah-bakhtawar@ubuntu:~$ _
```

File: *windows_ssh_identity.png*

Caption: Remote SSH prompt clearly showing the <github-username>@ubuntu format.